

Technology Stack

Date	6 July 2026
TeamID	
ProjectName	Plugging into the future:An Exploration of electricity consumption patterns
MaximumMarks	2Marks

TechnologyStackDetails

S.No	Architecture Component/Layer	Technology Chosen	Justification/Purpose
1	Frontend/Client-Side	<i>HTML,CSS, JavaScript</i>	Creates a responsive and intuitive user interface to display electricity consumption dashboards, charts, and input fields for data analysis.
2	Backend/Server-Side	<i>Python,Flask</i>	Handles user requests, processes historical power usage logs, executes the machine learning forecasting models, and returns energy consumption predictions.
3	Database/DataStorage	<i>CSVDataset/ SQLite</i>	Stores household or regional electricity consumption datasets, time-series logs, and model training data.
4	Cloud/Hosting/ Deployment	<i>IBMCloud</i>	Deploys the web application for online access and scalability.

5	VersionControl&CI /CD	<i>GitHub</i>	Manages source code and enables team collaboration.
6	Third-PartyAPIs/Other Tools	<i>Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn</i>	Used for energy data analysis, machine learning forecasting, preprocessing, and consumption pattern visualization.

